Example 2.6 Our monthly electricity bill contains a basic charge, that is independent of units consumed and a charge that depends on the units consumed. Let us say Electricity Board charges Rs.110 as basic charge and charges Rs. 4 for each unit we use. If a person wants to keep his electricity bill below Rs.250, then what should be his electricity usage?

Solution:
Let x denote the number of units used. Note that $x \geq 0$. Then, his electricity bill is Rs. $110 + 4x$. The person wants his bill to be below Rs.250. Let us solve the inequality $110 + 4x < 250$. Thus, $4x < 140$; which gives $0 \leq x < 35$.
The person should keep his usage below 35 units in order to keep his bill below Rs.250.

Example 2.7 Solve $3x - 5 \leq x + 1$ for x.

Solution:
We have $3x - 5 \leq x + 1$; which is equivalent to $2x \leq 6$. Hence we have $x \leq 3$; the solution set is $(-\infty, 3]$.

We can also solve the above inequality graphically.
Let us consider the graphs of $f(x) = 3x - 5$ and $g(x) = x + 1$ (See Figure 2.3). Now, find all the x-values for which the graph of f is below the graph of g.

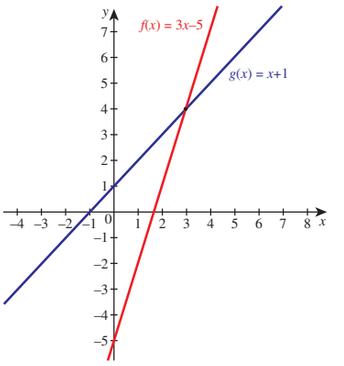


Figure 2.3

Example 2.8 Solve the following system of linear inequalities.
$3x - 9 \geq 0, \quad 4x - 10 \leq 6.$

Solution:
Note that $3x - 9 \geq 0$ implies $3x \geq 9$, by multiplying both sides by $1/3$ we get $x \geq 3$. Similarly, $4x - 10 \leq 6$ implies $4x \leq 16$ and hence $x \leq 4$.

So the solution set of $3x - 9 \geq 0, 4x - 10 \leq 6$ is the intersection of $[3, \infty)$ and $(-\infty, 4]$. Clearly, the intersection of these intervals give $[3, 4]$.